administrative tasks should be tracked in revision control, delivered with the application they support, and executed in the same environment as the application.

It’s usually a good idea to frame administrative tasks as small standalone services that run once and then are thrown away or as functions configured in a stateless platform to be triggered when certain events happen, or you can embed them in the application itself, activating them by calling a specific endpoint.

2.1.11 Port binding

Applications following the 15-Factor methodology should be self-contained and export their services via port binding. In production, there might be some routing services that translate requests from public endpoints to the internal port-bound services.

An application is self-contained if it doesn’t depend on an external server in the execution environment. A Java web application would probably run inside a server container like Tomcat, Jetty, or Undertow. A cloud native application, in contrast, would not require the environment to have a Tomcat server available; it would manage it itself as any other dependency. Spring Boot, for example, lets you use an embedded server: the application will contain the server rather than depending on one being available in the execution environment. One of the consequences of this approach is that there is always a one-to-one mapping between application and server, unlike the traditional method where multiple applications are deployed to the same server.

The services provided by the application are then exported via port binding. A web application would bind HTTP services to a specific port and